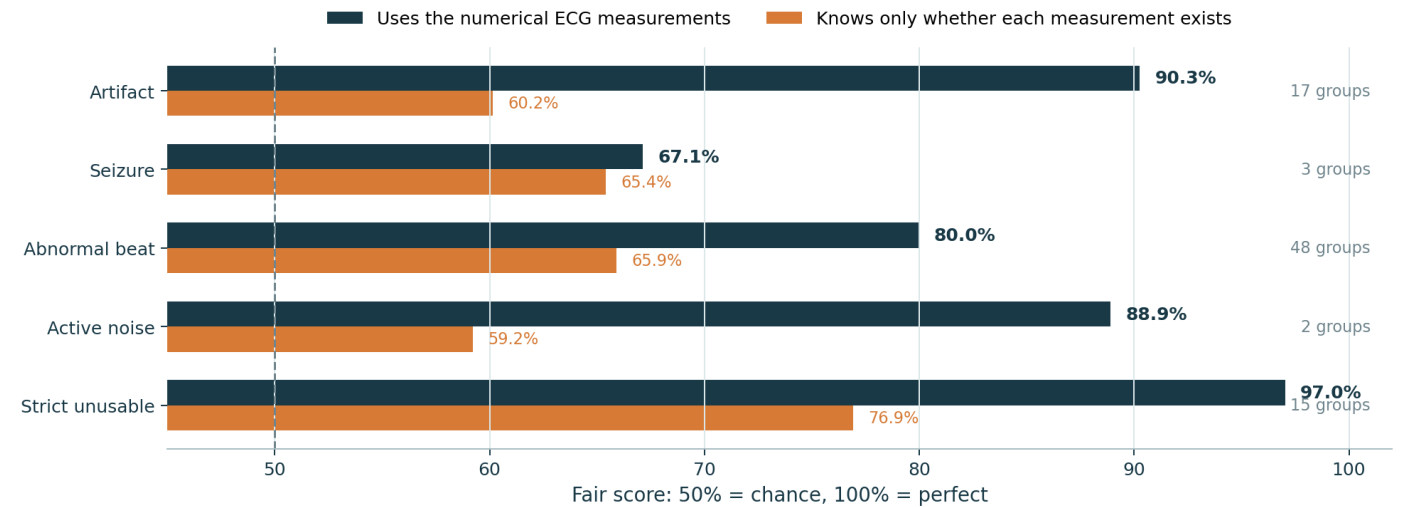


What the extracted ECG feature matrix can and cannot do

A four-page, seizure-focused interpretation for discussion with the PI

Clear conclusion. The 52 extracted features describe signal quality well and contain useful abnormal-beat information. They do *not* yet provide a reliable seizure detector. The seizure limitation is a combination of too few independent patients and a feature design centred on individual beats rather than changes over time.



Target	Fair score*	Found	Ignored	Alerts right	AUC	Groups	Plain conclusion
Artifact	90.3%	87.9%	93.7%	94.5%	95.4%	17	Strong internal result
Seizure	67.1%	63.0%	71.0%	19.7%	73.1%	3	Weak; too many false alerts
Abnormal beat	80.0%	69.9%	90.2%	64.3%	90.1%	48	Moderate; varies by patient
Active noise	88.9%	82.2%	95.7%	97.9%	95.0%	2	Strong-looking; not generalizable yet
Strict unusable	97.0%	92.7%	98.0%	88.0%	98.4%	15	Strong, but a narrow quality task

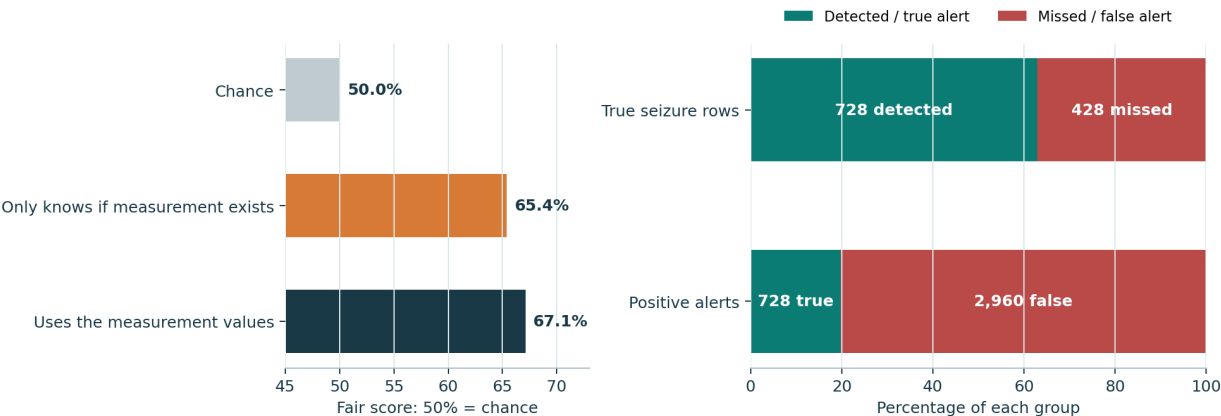
What every statistic means in ordinary language

Report term	Ask this clinical question	How to interpret it
Fair score* (group BA)	Does the model handle both positive and negative cases fairly?	Average of “found” and “ignored.” 50% is coin-flip performance. Each patient/source receives equal weight, so one long recording cannot dominate.
Found (sensitivity)	Out of 100 real seizures/artifacts, how many did it find?	Seizure: 63 found and 37 missed.
Ignored (specificity)	Out of 100 non-seizure/non-artifact cases, how many did it correctly leave alone?	Seizure: 71 ignored and 29 falsely flagged.
Alerts right (PPV)	Out of 100 alarms, how many were genuine?	Seizure: only about 20 genuine; about 80 false.
AUC	Does a genuine case usually receive a higher risk score than a non-case?	50% is random ranking; 100% is perfect ranking. It is <i>not</i> the percentage of correct alarms.

Do not over-interpret this. The navy and orange bars are two separate tests. Navy uses the numerical ECG measurements. Orange is deliberately denied those numbers and is told only whether each measurement exists. Page 2 gives a concrete example.

Seizure is the clear weakness

The numerical measurements barely improve on knowing only whether measurements were available



Clear conclusion. For seizure, the numerical ECG measurements achieved a 67.1% patient-weighted fair score. A control told only whether each measurement existed achieved 65.4%. The improvement is only 1.7 percentage points, so the current matrix has not learned a strong seizure signature.

Exactly how to read the seizure row

Number shown	What it means for these seizure predictions
67.1% fair score	Overall ability to separate seizure from non-seizure while giving both sides equal importance. It is better than the 50% chance level, but not strong.
63.0% found	Of 1,156 true seizure rows, 728 were detected and 428 were missed.
71.0% ignored	Of 10,190 non-seizure rows, 7,230 were correctly left alone and 2,960 were falsely flagged.
19.7% alerts right	The model produced 3,688 seizure alerts, but only 728 were correct. About four out of every five alerts were false.
73.1% AUC	Seizure rows often received somewhat higher scores than non-seizure rows when every possible alarm cutoff was considered. It does <i>not</i> mean 73.1% of alarms were correct.
3 independent groups	The nine recording files come from only three independent patient/source groups. Nine files do not equal nine independent patients.

Do not over-interpret this. Nothing in the seizure row is 90%. The 90.3% fair score belongs to artifact detection, not seizure detection.

Feature availability versus an actual ECG feature

Example	Availability-only test sees	Actual-feature test sees
Heart rate	“Was it calculated? Yes.”	“The heart rate is 112 beats/min.”
RR variability	“Was it calculated? Yes.”	“The variability is 38 ms.”
QRS width	“Was it calculated? No.”	No numerical measurement is available.

The availability test sees only a pattern such as **yes, yes, no**; it never sees 112 beats/min or 38 ms. It can still make guesses from dataset-specific missing-value patterns. The actual-feature test sees the numerical measurements extracted from the ECG, but not the raw waveform.

Why this remains a weak seizure result

There were too many false alarms, only three independent groups, and the matrix mainly describes isolated beats. Seizure effects are more likely to appear as sustained changes in heart rate, beat-to-beat variability and ECG shape relative to the patient’s baseline before, during and after an episode.

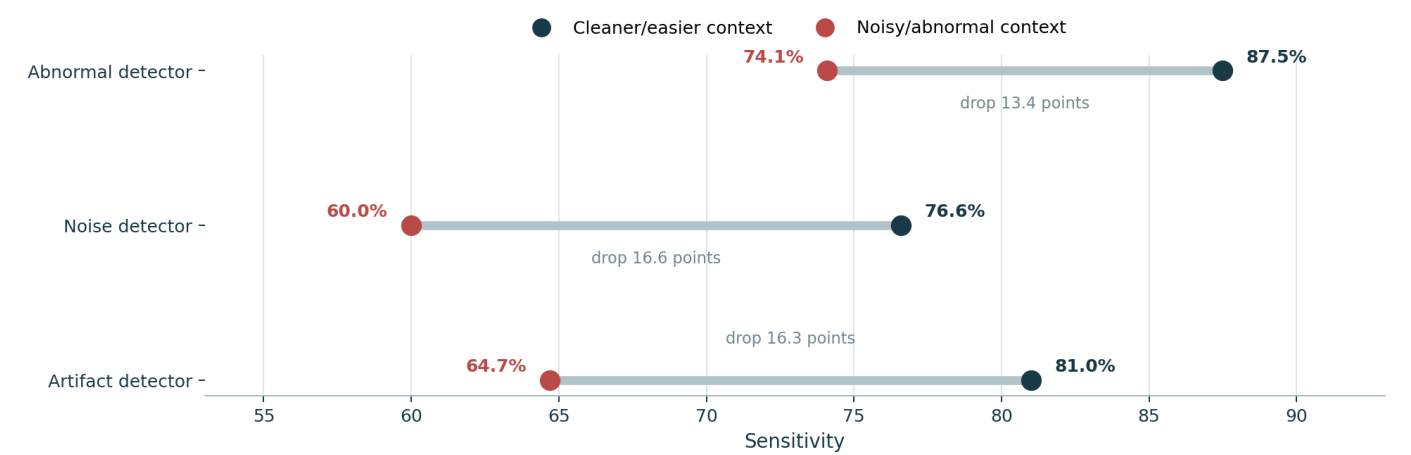
Concrete weak case: PN06_5

For record PN06_5, the model detected only 22 of 71 seizure rows and missed 49. It also alerted on 327 non-seizure rows. Its fair score was 49.3%, and only 6.3% of its alerts were correct: essentially chance performance for this record.

What this means for the next step. This does not mean ECG can never help detect seizures. It means this sample and beat-centred design are insufficient. Add more independent seizure patients, then describe changes over time relative to each patient’s usual ECG and evaluate whole seizure episodes.

Where the matrix fails under harder conditions

Abnormal morphology and noise reduce one another’s sensitivity; strong averages also contain weak records



Clear conclusion. When abnormal beats and noise coexist, detection becomes harder in both directions. Abnormal-beat sensitivity falls by 13.4 points during noise; noise sensitivity falls by 16.6 points on abnormal beats; artifact sensitivity falls by 16.3 points on abnormal beats.

How consistent was performance between patient/source groups?

Target	Mixed groups	Typical fair score	Middle half	Worst group	Interpretation
Artifact	15	95.0%	85.2–96.6%	63.0%	Usually strong; one important low group
Seizure	3	67.4%	66.3–68.1%	65.2%	Consistently weak; none reached 80%
Abnormal beat	40	82.3%	70.3–96.1%	49.5%	Broad evidence but large variation; six below 60%
Active noise	2	88.9%	88.7–89.1%	88.6%	Too few independent groups
Strict unusable	7	99.9%	99.5–100%	87.2%	Strong in seven mixed groups; eight others had one class

Weak records that the average hides

Task and record	Cases found	Fair score	What failed
Artifact: NSTDB 118e24	26.1%	60.4%	116 of 157 artifact rows were missed despite high overall artifact performance.
Active noise: NSTDB 118e24	42.0%	63.8%	91 of 157 noisy rows were missed.
Abnormal beat: MIT-BIH 207	35.0%	Not defined	67 of 103 abnormal anchors were missed; this selected window had no negatives.
Strict unusable: BUT 111001	74.5%	87.2%	47 of 184 unusable rows were missed even though there were no false unusable alarms.

What combinations could not be tested?

Noise/artifact and beat type coexist in 2,158 NSTDB rows, so the comparisons above are valid. The seizure records do not have synchronized noise schedules, expert AAMI beat classes or BUT-QDB quality labels. Therefore seizure plus noise, seizure plus abnormal beat and seizure plus artifact are **not evaluable** in this matrix. Unknown labels were excluded, not changed to zero.

Do not over-interpret this. A high average can hide false reassurance for one patient. The next model should be judged by patient-level distributions, the weakest groups and mixed-condition sensitivity, not only its overall average.

Quality-label detail: artifact combines degraded and unusable quality with active NSTDB noise. The strict unusable task is narrower: pure BUT-QDB class 1 versus class 3, with the intermediate class deliberately excluded.

What should change next

A seizure-specific data and feature redesign, followed by a locked patient-level test

Step	Evidence requiring action	Change	How to know it helped
1	Only three seizure groups	Add substantially more independent seizure patients, seizure episodes and difficult non-seizure controls.	Performance remains stable for entirely unseen patients and sources.
2	Rhythm and beat-to-beat timing performed best	Build 30-second, 1-minute, 2-minute and 5-minute summaries: heart-rate acceleration, changing beat-to-beat variability, rhythm irregularity, abnormal-beat burden, ECG-shape trends and change from the patient's usual level.	The new over-time features beat the current features when tested on exactly the same unseen patients.
3	The yes/no availability test already scored 65.4%	Use the same measurement panel across datasets; rerun without the yes/no availability clues; test rows with a complete common panel and test on an entirely different dataset.	Numerical ECG values clearly beat the yes/no control for each held-out patient, not just on average.
4	Individual-row alerts produced 2,960 false alarms	Judge whole seizure episodes; require a pattern to persist; choose the alarm cutoff carefully; allow "uncertain" when signal quality is poor.	More seizure episodes found, fewer false alarms per hour and acceptable time to detection.
5	The winner was chosen from 375 tested options	Make every model and feature choice using training patients only. Then lock the complete method before one final test on untouched external patients.	The locked method repeats its performance in a new patient group, device and clinical-site mix.

The next experiment I would run

1. Keep the current best seizure model—rhythm/beat-timing features with a linear SVM—as the fixed baseline. Do not tune it again using the final test patients.
2. Build three versions: **A** current features; **B** the same measurements available across every dataset; and **C** new features describing changes across 30 seconds to 5 minutes relative to each patient's usual level.
3. Compare A, B and C on exactly the same held-out patients. Choose the model, features and alarm cutoff using training patients only.
4. Report the percentage of seizure episodes found, false alarms per hour, time to detection, percentage of alarms correct, fair score for each patient and improvement over the yes/no availability control.
5. Keep PN06_5 and the other weak records as fixed challenge cases. Improvement is credible only if the weakest patients improve, not merely the average.

Recommended cascade

Raw ECG → quality and uncertainty → beat detection → temporal seizure features → episode score → alert / no alert / uncertain

What can be said to the PI now

Supported by this audit

- The matrix was rebuilt exactly and evaluated without patient/source overlap.
- The features are useful for signal quality and contain moderate abnormal-beat information.
- The present seizure representation is inadequate and its main limitations are now identifiable.

Not supported yet

- A clinically reliable seizure detector.
- Generalization of the noise result beyond two source patients.
- Seizure performance during artifact or abnormal beats, because joint labels are absent.
- Autonomous diagnosis or signal rejection.

Clear conclusion. The work has produced a reliable internal audit and a clear redesign target. The most productive next step is not another generic classifier: it is more independent seizure data plus patient-normalized temporal features and episode-level validation.